

Immunogenicity and safety of a 14-valent pneumococcal polysaccharide conjugate vaccine (PNEUBEVAX 14™) administered to 6–8 weeks old healthy Indian Infants: A single blind, randomized, active-controlled, Phase-III study

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ABSTRACT

Background: Introduction of pneumococcal conjugate vaccines (PCVs) reduced the number of cases of pneumococcal disease (PD). However, there is an increase in clinical and economic burden of PD from serotypes that are not part of the existing pneumococcal vaccines, particularly impacting pediatric and elder population. In addition, the regions where the PCV is not available, the disease burden remains high. In this study, immunogenicity and safety of the BE's 14-valent PCV (PNEUBEVAX 14™; BE-PCV-14) containing two additional epidemiologically important serotypes (22F and 33F) was evaluated in infants in comparison to licensed vaccine, Prevenar-13 (PCV-13).

Methods: This is a pivotal phase-3 single blind randomized active-controlled study conducted at 12 sites across India in 6–8 weeks old healthy infants at 6–10–14 weeks dosing schedule to assess immunogenic non-inferiority and safety of a candidate BE-PCV-14. In total, 1290 infants were equally randomized to receive either BE-PCV-14 or PCV-13. Solicited local reactions and systemic events, adverse events (AEs), serious AEs (SAEs), and medically attended AEs (MAAEs) were recorded. Immunogenicity was assessed by measuring anti-PnCPS (anti-pneumococcal capsular polysaccharide) IgG concentration and functional antibody titers through opsonophagocytic activity (OPA), one month after completing three dose schedule. Cross protection to serotype 6A offered by serotype 6B was also assessed in this study.

Findings: The safety profile of BE-PCV-14 was comparable to PCV-13 vaccine. Majority of reported AEs were mild in nature. No severe or serious AEs were reported in both the treatment groups. For the twelve common serotypes and for the additional serotypes (22F and 33F) in BE-PCV-14, NI criteria was demonstrated as defined by WHO TRS-977. Primary immunogenicity endpoint was met in terms of IgG immune responses for all 14 serotypes of BE-PCV-14. Moreover, a significant proportion of subjects (69%) seroconverted against serotype 6A, even though this antigen was not present in BE-PCV-14. This indicates that serotype 6B of BE-PCV-14 cross protects serotype 6A. BE-PCV-14 also elicited comparable serotype specific functional OPA immune responses to all the serotypes common to PCV-13.

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Interpretations: BE-PCV-14 was found to be safe and induced robust and functional serotype specific immune responses to all 14 serotypes. It also elicited cross protective immune response against serotype 6B. These findings suggest that BE-PCV-14 can be safely administered to infants and achieve protection against pneumococcal disease caused by serotypes covered in the vaccine.

The study was prospectively registered with clinical trial registry of India – CTRI/2020/02/023129.

1. Introduction

Streptococcus pneumoniae is a gram positive bacterium, causing morbidity and mortality globally in children under 5 years, adults above 50 years and in individuals with comorbidities. It manifests a variety of invasive or noninvasive diseases such as pneumonia, otitis media, nasosinusitis, bacteremia and meningitis [1,2]. Most of the deaths are caused by invasive pneumococcal disease (IPD) with reported >300,000 deaths worldwide in children under 5 years every year due to pneumonia or meningitis [3]. In developing countries, case fatality rates may be as high as 50 % for meningitis and up to 20 % for septicemia. As per a global population-based model in 2015 and as per WHO data, case fatality rates under 5 years' is the highest in India followed by Pakistan, Bangladesh, Sri Lanka and Indonesia with reported incidences over 200/100,000 population [4,5]. Although there are more than 95 identified serotypes of *S. pneumoniae*, few serotypes are responsible for causing the disease [6]. In addition, in more than 30 % of cases, pneumococcal bacteria are resistant to one or more antibiotics [7].

A 23-valent, unconjugated pneumococcal polysaccharide vaccine (PPSV23; Pneumovax® 23, Merck and Co., Inc., NJ) provides only transient protection as it elicits T-cell independent immune responses [8]. Pneumococcal polysaccharide-protein conjugate vaccines (PCVs) 7- and 13-valent (PCV-7; Prevenar 7®; PCV-13; Prevenar 13®; Pfizer, New York), were able to induce T-cell dependent immune response against respective serotypes and generate robust functional antibody responses providing long-term protection against IPD across different age groups [9–11]. Introduction of PCVs substantially reduced the burden of IPD worldwide in all age groups [12]. Despite these reductions, one-fourth to one-third of the *S. pneumoniae* serotypes causing IPD are not covered by currently available PCVs and these serotypes bring about a significant unmet medical need [13–15]. A study by Balsells, et al. [16] reported that 42.2 % of childhood IPD cases were caused by serotypes, that are not included in the existing pneumococcal vaccines. So, it is important to include additional serotypes to minimize the burden of IPD. Biological E Limited developed a PCV covering 14 different serotypes (PNEUBEVAX 14™) including two serotypes (22F and 33F) that are not present in PCV-13. These two serotypes were reported to be among top 11 predominant vaccine serotypes, not part of the current pneumococcal vaccines, which accounted for 22 % of invasive pneumococcal disease cases in India under-five years [17,18]. A recent study [19], also reported that 22F and 33F are the most prevalent serotypes reported in IPD [20]. Moreover, these two serotypes are among the first in serotype replacement activity after introduction of PCV-7 and PCV-13 vaccines [19,20]. A developing country such as India can see higher benefit with PCV containing new serotypes reported for disease burden..

Safety and immunogenicity of BE-PCV-14 was established in phase-1 and phase-2 studies in healthy adults and toddlers respectively (CTRI/2017/06/008759 and CTRI/2017/11/010387). Here we report the immunogenicity and safety of BE-PCV-14 in 6–8 weeks old healthy infants (n = 1290) in 6–10–14 weeks dosing schedule demonstrating the immunogenic non-inferiority with the comparator PCV-13.

2. Methods

2.1. Study design and participants

The phase-3, two arm, single blind, randomized, active-controlled study was conducted between February 2020 and March 2022 in India

across 12 sites. The study was designed to evaluate the immunogenic non-inferiority and safety of a BE-PCV-14 administered to 6–8 weeks old healthy infants in 6–10–14 weeks dosing schedule. Participants were equally randomized (1:1) to receive either BE-PCV-14 (test) or PCV-13 (Prevenar 13®, comparator). In total, 1290 subjects were enrolled into the study (n = 645 in BE-PCV-14 arm and n = 645 in PCV-13 arm). A dose of 0.5 mL of the study vaccine (BE-PCV-14 or PCV-13) was administered intramuscularly in a three dose vaccination schedule (3 + 0) with a 4-weeks interval.

All enrolled subjects (N = 1290) were healthy infants with ≥3300 g weight at the time of screening. None of the subjects received any licensed or investigational pneumococcal vaccine prior to enrollment. Other permitted vaccines included single dose of Bacillus Calmette-Guérin

(BCG), hepatitis B, Diphtheria, Tetanus, whole cell Pertussis, Hepatitis B and Haemophilus influenza type b (DTwP-HepB-Hib; liquid pentavalent), polio and rotavirus vaccines. Informed consent was obtained from subjects' parent(s) prior to performing any study specific procedure. Exclusion criteria used were evidence of previous *S. pneumoniae* infection, use of any investigational or non-registered product (drug or vaccine) –30 days prior to study vaccine administration, history of any neurological disorders, meningitis, or seizures, family history of congenital or hereditary immunodeficiency, history of allergic disease or history of a serious reaction to any prior vaccination or known hypersensitivity likely to be exacerbated by any component of the study vaccines. Complete list of eligibility criteria provided as [supplementary information](#).

The study was conducted in accordance with the principles defined in the Declaration of Helsinki, International Conference on Harmonization guidelines (Good Clinical Practices), and the new drug and clinical trial rules, 2019 [21]. The investigational review board or ethics committee at each study site approved the protocol.

2.2. Randomization and masking

Eligible subjects were randomized to receive either BE-PCV-14 or PCV-13 in 1:1 ratio and randomization numbers were assigned using an interactive web-based randomization system (IWRs). Randomization sequence was generated using SAS software (SAS Institute Inc, 100 SAS Campus Drive, Cary, NC 27513-2414, USA).

2.3. Procedure

Screening, enrolment, and first dose of primary vaccination was done when infants were 6–8 weeks old (visit-1; day 0). Second and third doses of primary vaccination (visit-2; day 28 and visit-3; day 56) and a final follow up immunogenicity visit (visit-4, day 86) took place at four weeks' interval.

An additional time window of +7 days for the second, third, and the fourth visit was permitted to ensure subject's visit compliance.

Each 0.5 mL of single human dose of BE-PCV-14 (PNEUBEVAX 14™) contained polysaccharide-protein conjugate antigens 3 µg of serotype 1 and 2.2 µg of each of serotypes 3, 4, 5, 7F, 9V, 14, 18C, 19A, 19F, 22F, 23F and 33F and 4.4 µg of serotype 6B polysaccharides conjugated to about 35 µg of non-toxic diphtheria toxin cross-reacting material 197 (CRM₁₉₇) protein and adsorbed onto ≤0.75 mg of aluminium phosphate (Al+++). The 13-valent Pneumococcal polysaccharide conjugate vaccine "PCV-13" (control vaccine) contained 2.2 µg of each of serotypes 1,

3, 4, 5, 6A, 7F, 9V, 14, 18C, 19A, 19F, 23F saccharides, and 4.4 µg of 6B saccharide, 34 µg CRM₁₉₇ carrier protein, and 125 µg aluminum as aluminum phosphate adjuvant. Both vaccines were administered in the anterolateral aspect of thigh.

Approximately 5.0 mL of blood was collected at visit-1 (day 0) and again at visit-4 (day 86). Anti-pneumococcal capsular polysaccharide (PnCPS) IgG antibody concentration estimation against each of the 14 vaccine serotypes was performed as per the World Health Organization (WHO) reference enzyme linked immunosorbent (ELISA) assay [22] with a minor modification. Cell wall polysaccharide (CWPS) multi was used instead of CWPS and 22F pneumococcal polysaccharide (PnCPS) for pre-absorption to neutralize non-specific antibodies and other common contaminants present in the PnCPS coating antigens. Functional antibody responses against all 14 serotypes were assessed by opsonophagocytic activity (OPA) assay, which was performed at WHO pneumococcal serology reference laboratory (UCL Great Ormond Street, Institute of Child Health, Immuno-biology Section, United Kingdom).

Each vaccinated infant was observed for at least 60 min after vaccination to evaluate and treat for any immediate adverse reactions. At each vaccination visit, a diary card was provided to the subject's parent(s)/legally acceptable representative (LAR) to record any solicited local/general AE occurring after vaccination for 7 consecutive days (day 0 - day 6). List of solicited local and systemic AEs are provided as [supplementary information](#).

Unsolicited AEs, serious adverse events (SAEs), medically attended adverse events (MAAEs) were recorded until the end of the study. Severity of all AEs was graded as per the common terminology criteria for adverse events (CTCAE V5.0) or division of AIDS (DAIDS) table version 2.0. Brighton collaboration scale was taken for severity grading of fever. All AEs were assessed for relatedness to study vaccine by the investigator.

2.4. Outcomes

Primary outcome of the study was to demonstrate serotype specific immunogenic non-inferiority (NI) of BE-PCV-14 to licensed PCV-13 vaccine one month after the completion of primary vaccination series against 12 common serotypes (1, 3, 4, 5, 6B, 7F, 9V, 14, 18C, 19A, 19F, and 23F) in terms of number of subjects seroconverted and the ratio of IgG geometric mean concentrations (GMCs). Immunogenic NI for new serotypes 22F and 33F in BE-PCV-14 was demonstrated by comparing immunoglobulin G (IgG) concentrations induced by serotypes 22F and 33F of BE-PCV-14 with one of the PCV-13 serotypes that showed the lowest sero-response. Sero-response or seroconversion rate was defined as proportion of vaccinated infants with serotype specific IgG antibody concentrations of at least ≥ 0.35 µg/ml [23].

One of the secondary objectives of the study was to assess the immune response in terms of serotype specific opsonophagocytic activity (OPA) in a randomly chosen subset of subjects from the BE-PCV-14 group compared with an equal number of subjects randomly chosen from PCV-13 vaccinated group. Other secondary outcome was to evaluate the safety of BE-PCV-14 in comparison with PCV-13 measured by the incidence rates of reactogenicity events (local reactions and systemic events) and adverse events (AEs).

2.5. Statistical analyses

2.5.1. Sample size determination

Sample size estimation was based on the 2-sided 95 % confidence interval (CI) for the difference in proportion of subjects achieving a serotype specific pneumococcal IgG concentration ≥ 0.35 µg/mL in each vaccine group. After study completion, a two-sided 95 % confidence interval for the true difference between the two proportions was constructed. The lower bound of the confidence interval was to lie entirely on the positive side of the -10 % points margin to demonstrate non-inferiority.

Data from the phase-II safety and immunogenicity study in 12–23 months old children conducted in India (CTRI/2017/11/010387) was used to support sample size estimation. The power was computed for an anticipated test vaccine group proportion of 92.33 % within the non-inferiority margin set. The significance level of the test was targeted at 0.025. Accordingly, sample sizes of 581 in each group was estimated to achieve 90 % power and to detect a non-inferiority margin difference between the group proportions of -0.10. An anticipated, not more than 10 % dropout rate was built-in to the sample size. Therefore, the overall sample size required was 645 infants per group to obtain 581 evaluable infants per group at the end of the study.

2.6. Statistical analyses

All demographic and baseline characteristics of both the groups were presented using intend-to treat (ITT) population, defined as all subjects randomized into the study. The data was analyzed by summary statistics. All subjects who received at least one dose of the study vaccine were included in the safety analysis and the data was summarized descriptively. For continuous variables *n*, *mean*, *standard deviation*, *median* and *range* (minimum, maximum) have been presented. For categorical data, frequencies were computed. All reported adverse events during the entire study period were summarized by calculating frequencies and listed per subject including severity and relationship (causality).

The number and percentage of subjects with AEs were presented by system organ class and preferred term. All medically attended AEs reported during the study were listed and analyzed for expectedness and causality. Two-sided 95 % exact confidence intervals were calculated for all the occurrence rates of reported AEs and SAEs during the study. All AEs were coded using the Medical Dictionary for Regulatory Activities (MedDRA™; version 25.1) coding dictionary and concomitant medications have been coded using the WHO Drug Dictionary, March 2022.

Immunogenicity assessment was based on the per-protocol population defined as the vaccinated subjects whose blood samples were available for immunogenicity analysis at both baseline and day 86 (visit-4). Non-inferiority of serotype specific immune responses induced by BV-PCV-14 against PCV-13 was assessed by one of the below two criteria as per the WHO defined criteria [23].

1. By measuring difference in proportion of subjects seroconverted to achieve serotype specific anti-PnCPS IgG concentration ≥ 0.35 µg/ml between BE-PCV-14 and PCV-13. The lower limit of 95 % CI for the difference in proportion of subjects seroconverted between BE-PCV-14 and PCV-13 should be above the minus 10 % points.
2. By assessing the IgG GMC ratio (BE-PCV-14/PCV-13) along with its two sided 95 % confidence interval. Non-inferiority was considered achieved if the ratio of lower limit of two sided 95 % CI (test /control group) was above the pre-specified margin of 0.5.

For two new serotypes 22F and 33F which were not part of PCV-13, non-inferiority was assessed by comparing with any serotype in the PCV-13 group that achieved the lowest percentage of ≥ 0.35 µg/ml IgG GMC. For serotype 6A, which was not in BE-PCV-14 but present in PCV-13, a ≥ 2 -fold increase in the 6A specific geometric mean concentrations (GMCs) above pre-vaccination levels was assessed as an exploratory outcome. Proportion of subjects achieving ≥ 4 -fold rise in anti-PnCPS IgG antibody concentrations from baseline and geometric mean fold rise (GMFR) along with their 2-sided 95 % CI against each of the vaccine serotype was calculated.

As a secondary outcome, proportion of subjects achieving a serotype specific OPA titre $\geq 1:8$ post vaccination was assessed along with 2-sided 95 % confidence. Serotype specific OPA geometric mean titres (GMT) along with their ratios and GMFR against 12 common serotypes in both vaccine groups was evaluated.

A reverse cumulative distribution (RCD) curve of anti-PnCPS antibody concentrations and OPA GMT titers by serotype of BE-PCV-14 and

PCV-13 were plotted for all common vaccine serotypes in both groups.

3. Results

A total of 1290 subjects were screened and randomly assigned to receive either BE-PCV-14 ($n = 645$) or PCV-13 ($n = 645$) study vaccine. Out of 1290 subjects, 1267 (98.2 %) subjects (641 in BE-PCV-14 arm and 626 in PCV-13 arm) completed all the visit specific procedures and were included in the immunogenicity (anti-PnCPS IgG antibody concentration) analysis. In total, 22 (1.7 %) subjects were dropouts. Subject disposition with reason for dropout from the study was presented in Fig. 1. Safety analyses were performed in all enrolled subjects ($n = 1290$) whereas immunogenicity (per-protocol population) was completed in 1267 subjects. One subject was excluded from per-protocol population due to insufficient serum sample and the rest ($n = 22$) were dropouts. Demographics of intention-to-treat population were presented in Table 1. Age of the subjects and male to female ratio was comparable in both BE-PCV-14 and PCV-13 arms. Summary of other eligible vaccinations received by study subjects was presented in Supplementary Table 1.

The proportion of subjects with anti-PnCPS IgG antibody concentrations equal to or above the protective threshold $\geq 0.35 \mu\text{g/mL}$ (primary outcome) at day 86 in BE-PCV-14 ($n = 641$) group was in the range of 76.3 % (serotype 6B) to 99.7 % (serotype 14) and in the PCV-13 ($n = 626$) group, the response rate was in the range of 77 % (serotype 3) to 100 % (serotype 14) for 12 common serotypes. The lower limit of 95 % CI for the difference in proportion of subjects seroconverted between BE-PCV-14 and PCV-13 was above the minus 10 % points for all the 12 common serotypes (serotype 1, 3, 4, 5, 6B, 7F, 9V, 14, 18C, 19A, 19F and 23F) and all of them met pre-defined NI criteria. The distributions in serotype specific IgG concentrations for 12 common serotypes in both vaccine groups were similar, as shown by the reverse cumulative distribution curves (Fig. 3).

Seroresponse rates for serotype 22F (94.1 %) and 33F (73.2 %) that are present only in BE-PCV-14 group were compared with the lowest performing serotype 3 (77.0 %) in PCV-13 group. Defined NI criteria was met for both serotypes 22F and 33F (Fig. 2a).

For IgG geometric mean concentrations, NI criteria was achieved for all 12 common serotypes and for additional 22F and 33F serotypes (Fig. 2b). The lower bound of the 95 % CI for GMC ratio was in the range of 0.67 (serotype 1) to 1.10 (serotype 3) for 12 common serotypes present in BE-PCV-14 and PCV-13. For serotype 3 and serotype 14, lower

Table 1

Demographic characteristics of study participants.

Parameter/ Statistics/Category	BE-PCV-14 (N = 645)	PCV-13 (N = 645)	Overall (N = 1290)
Age (Days)			
N1	645	645	1290
Mean	48.3	48.3	48.3
SD	3.75	3.69	3.72
Median	48.0	48.0	48.0
Range (Min: Max)	(42.0:56.0)	(42.0:56.0)	(42.0:56.0)
Gender, N1 (%)			
Male	342 (53.0 %)	365 (56.6 %)	707 (54.8 %)
Female	303 (47.0 %)	280 (43.4 %)	583 (45.2 %)
Height (Cms)			
N1	645	645	1290
Mean	52.0	52.0	52.0
SD	3.76	4.03	3.89
Median	51.4	51.2	51.3
Range (Min: Max)	(40.0:69.0)	(41.0:72.0)	(40.0:72.0)
Weight (Gms)			
N1	645	645	1290
Mean	4162.9	4165.7	4164.3
SD	526.85	526.44	526.44
Median	4100.0	4100.0	4100.0
Range (Min: Max)	(3300.0:6000.0)	(3300.0:6300.0)	(3300.0:6300.0)
Nationality, N1 (%)			
Indian	645 (100.0 %)	645 (100.0 %)	1290 (100.0 %)
BMI			
N1	645	645	1290
Mean	15.5	15.5	15.5
SD	2.19	2.31	2.25
Median	15.6	15.7	15.7
Range (Min: Max)	(8.0:25.0)	(8.0:26.5)	(8.0:26.5)

Note: Percentages were calculated using respective column header count as denominator.

N1: Subject Count, N: Sample Size.

bound 95 % CI was 1.10 and 1.01 respectively, indicating that BE-PCV-14 induces strong immune response compared to PCV-13. Serotypes 22F and 33F, which are not present in PCV-13, the lower bound of the 95 % CI for GMC ratio was 5.26 and 1.82 respectively (Fig. 2b), when compared to the lowest performing serotype in PCV-13 (serotype 3).

The proportion of subjects achieving a serotype specific OPA titer $\geq$

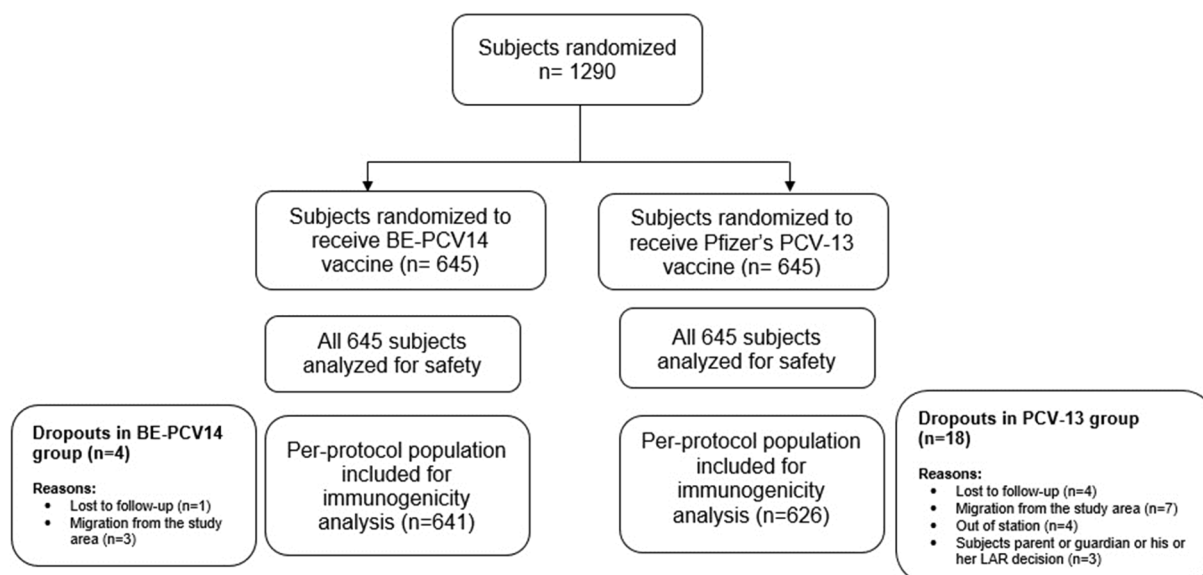


Fig. 1. Subject disposition. n = number; BE-PCV-14: Biological E's 14-valent PCV; PCV-13: Pfizer's 13-valent PCV.

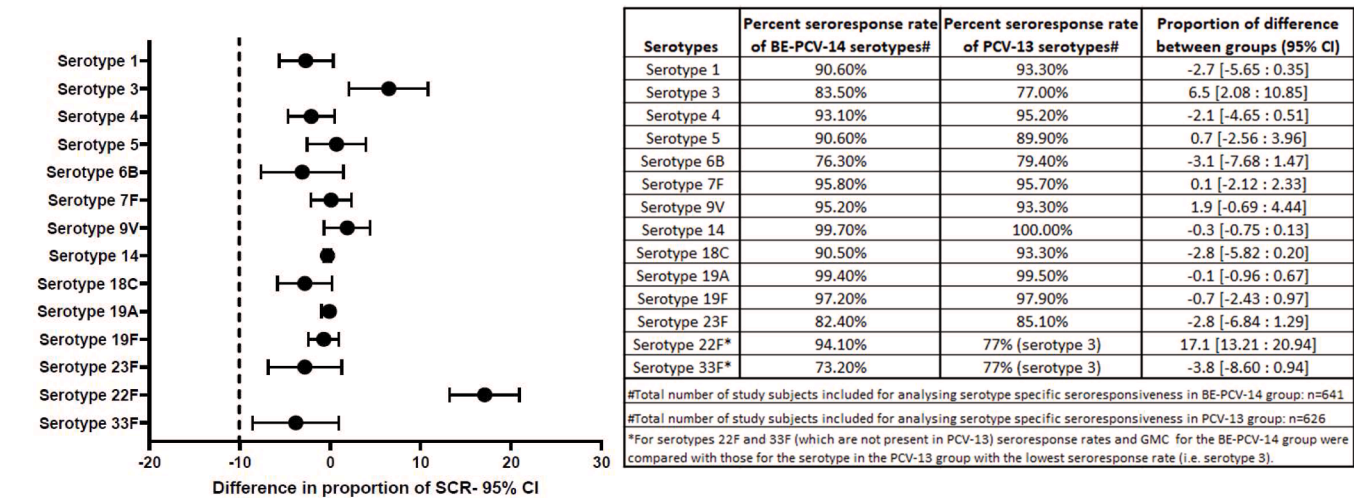


Fig. 2a. Difference in proportion of seroconversion rates with 95% confidence interval. CI: Confidence Interval; IgG: Immunoglobulin G; GMC: geometric mean concentrations; %: Percentage.

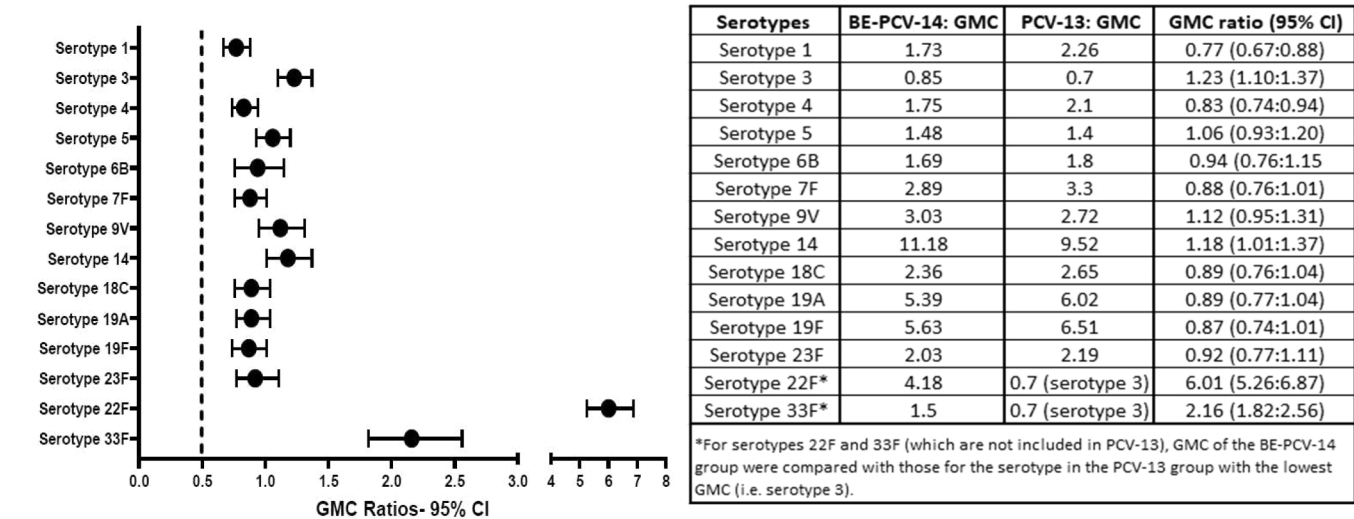


Fig. 2b. Geometric Mean Concentration ratios with 95% confidence intervals. CI: Confidence Interval; IgG: Immunoglobulin G; GMC: geometric mean concentrations; %: Percentage.

LLOQ ($\geq 1:8$) at day 86 were in the range of 63.64 % (serotype 1) to 98.43 % (serotype 22F) in BE-PCV-14 group and 64.43 % (serotype 1) to 98.46 % (serotype 7F and serotype 23F) in PCV-13 group. Overall, the OPA GMTs were comparable among all common serotypes in both vaccinated groups (Table 2).

The additional serotypes 22F and 33F in BE-PCV-14 vaccine had 15.16 and 16.91-fold rise in OPA GMTs respectively at day 86 compared to pre-vaccination levels, signifying that both the additional serotypes can induce high levels of protective functional antibodies in vaccinated infants (Table 2). The proportion of subjects achieving ≥ 4 -fold increase in anti-PnCPS IgG antibody concentrations at day 86 from baseline is presented in Table 3.

BE-PCV-14 vaccine has serotype 6B and no serotype 6A. In this study, we tested the cross protection against serotype 6A, assessed by a ≥ 2 -fold increase in the 6A specific IgG antibody concentrations and the OPA antibody titers above the pre-vaccination levels. There was 2.4-fold (pre-vaccination GMC: 0.45 and post-vaccination GMC: 1.08) and 7.64-fold (OPA GMT were 16.59 and 134.50 for pre- and post-vaccination, respectively) increase in serotype 6A specific IgG and OPA antibody titers respectively (Table 4). Our data on serotype 6A specific 69.7 % seroconversion and 2.4-fold rise in GMC in BE-PCV-14 vaccinated

infants along with functional antibody titers (OPA) GMFR of 7.64 fold strongly suggests that the serotype 6B confers significant immunological cross protection against serotype 6A.

The current study demonstrated that BE's 14-valent pneumococcal polysaccharide conjugate vaccine, administered in a 3-dose schedule (6–10–14 weeks), was safe, immunogenic and non-inferior to PCV-13 for the 12 common serotypes.

Among all vaccinated infants ($n = 1290$), 175/645 (27.1 %) subjects and 178/645 (27.6 %) subjects reported at least one adverse event in BE-PCV-14 and PCV-13 groups respectively. All the reported adverse events were generally mild and only very few cases were moderate in their intensity. Majority of the reported adverse events were solicited in nature and related to the study vaccine. Overview of the reported AEs including solicited and unsolicited AEs were presented in Table 5. Summary of AEs by system organ class (SOC) and preferred term (PT) were shown in Table 6. No severe or serious AEs were reported in either of the vaccine groups. Overview of AEs by severity grade and causality to the study vaccine were presented in Table 7. Summary of local, solicited systemic and unsolicited AEs by SOC and PT were presented in Supplementary Tables 2–4. No clinically significant changes overtime were noted in the vital signs. The AEs observed and physical

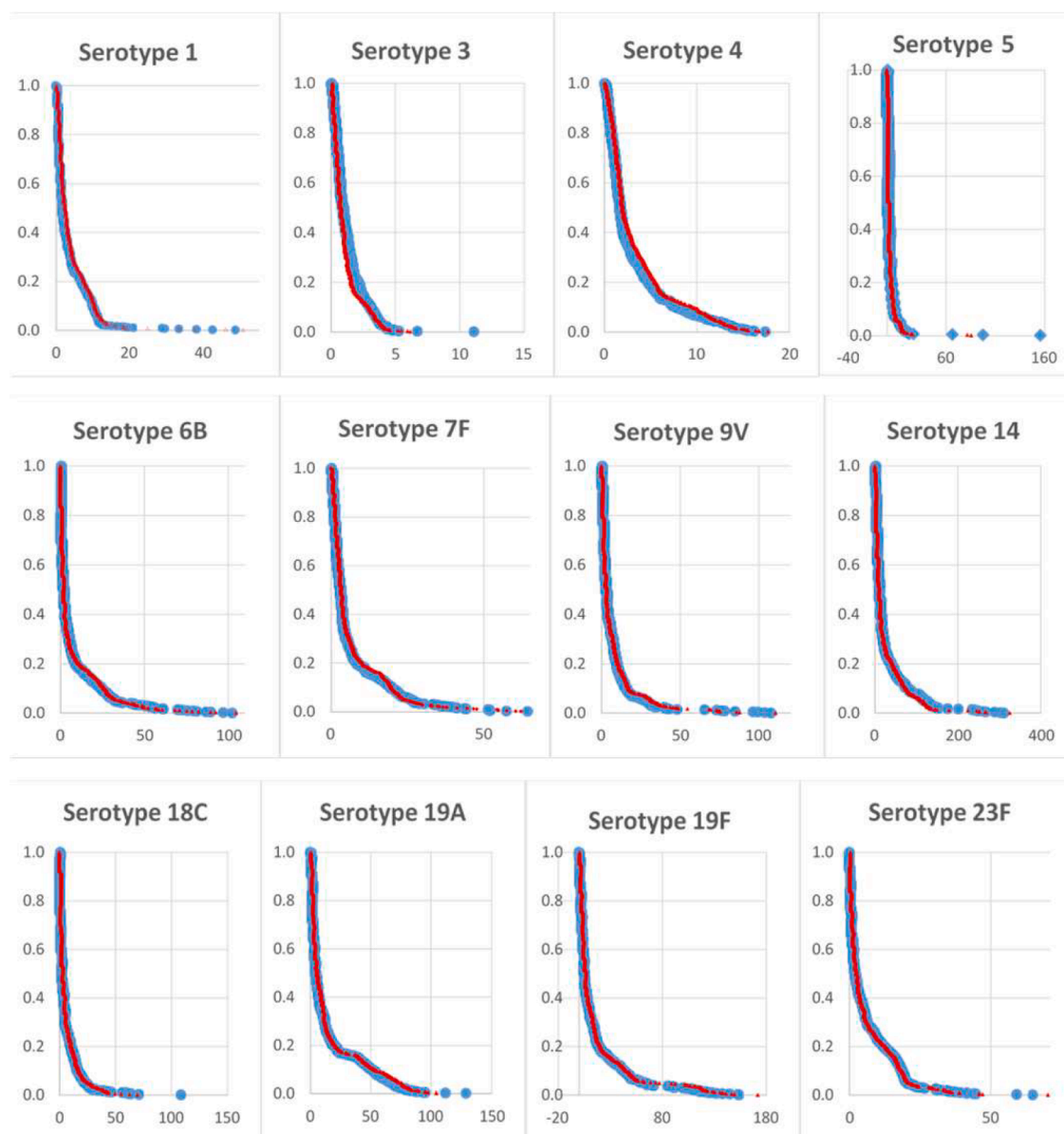


Fig. 3. Reverse cumulative distribution (RCD) curves of anti-PnCPS IgG antibody concentrations for 12 common serotypes between BE-PCV-14 and PCV-13. RCD: Reverse cumulative distribution; PnCPS IgG: Pneumococcal capsular polysaccharide specific immunoglobulin G; GMC: geometric mean concentrations; %: Percentage.

Table 2

Summary of serotype specific OPA titer $\geq$ LLOQ ($\geq 1:8$) in terms of sero-conversion rates, GMTs and fold rise.

Serotype	OPA sero-conversion rate		OPA GMTs		Percentage of subjects with at least 4 fold rise in OPA GMTs post vaccination		GMFR	
	BE-PCV-14	PCV-13	BE-PCV-14	PCV-13	BE-PCV-14	PCV-13	BE-PCV-14	PCV-13
Serotype 1	63.64	64.43	23.21	19.94	45.99	42.78	3.81	3.32
Serotype 3	71.81	79.58	59.6	63.01	62.77	61.26	9.97	9.86
Serotype 4	93.68	91.75	426.67	626.04	73.16	73.71	25.92	52.55
Serotype 5	80.73	84.1	66.47	78.17	60.94	66.67	8.95	11.61
Serotype 6B	88.48	88.21	990.66	918.24	70.68	68.72	26.32	27.9
Serotype 7F	97.4	98.46	1023.16	1156.95	71.35	76.92	14.56	19.97
Serotype 9V	97.92	97.94	856.81	746.44	78.65	72.16	24.18	27.22
Serotype 14	95.81	97.44	1951.14	1272.2	61.78	61.03	16.85	13.37
Serotype 18C	93.23	94.36	460.7	561.01	63.02	64.1	15.89	23.39
Serotype 19A	95.83	95.9	394.65	451.18	68.75	76.41	17.57	22.2
Serotype 19F	94.79	92.31	604.3	439.94	83.33	79.49	34.88	26.39
Serotype 23F	97.37	98.46	768.79	995.13	68.42	73.85	16.77	27.8
Serotype 22F	98.43	NA	691.6	NA	61.26	NA	15.16	NA
Serotype 33F	94.18	NA	1859.67	NA	60.85	NA	16.91	NA

Table 3
Proportion of subjects achieving ≥ 4 -fold increase in anti-PnCPS IgG antibody concentrations.

Anti-PnCPS IgG against each of the vaccine serotype post vaccination (Day 86)	BE-PCV14 (N = 641) N1 (%) [95 % CI]	PCV-13 (N = 626) N1 (%) [95 % CI]
Serotype 1	394 (61.5 %) [57.58:65.25]	441 (70.4 %) [66.70:74.00]
Serotype 3	299 (46.6 %) [42.73:50.59]	253 (40.4 %) [36.54:44.38]
Serotype 4	241 (37.6 %) [33.83:41.48]	294 (47.0 %) [43.00:50.96]
Serotype 5	388 (60.5 %) [56.63:64.34]	358 (57.2 %) [53.21:61.10]
Serotype 6B	341 (53.2 %) [49.25:57.12]	324 (51.8 %) [47.76:55.74]
Serotype 7F	455 (71.0 %) [67.30:74.47]	463 (74.0 %) [70.34:77.36]
Serotype 9V	422 (65.8 %) [62.02:69.50]	390 (62.3 %) [58.37:66.11]
Serotype 14	172 (26.8 %) [23.44:30.44]	157 (25.1 %) [21.73:28.67]
Serotype 18C	230 (35.9 %) [32.16:39.73]	243 (38.8 %) [34.98:42.76]
Serotype 19A	212 (33.1 %) [29.44:36.87]	230 (36.7 %) [32.95:40.65]
Serotype 19F	410 (64.0 %) [60.11:67.69]	419 (66.9 %) [63.09:70.61]
Serotype 23F	362 (56.5 %) [52.54:60.35]	371 (59.3 %) [55.30:63.14]

Note: Percentage was calculated using respective header count as denominator 95% CI for was calculated using Clopper Pearson Method.

Table 4
Serotype-6A specific IgG and OPA immune responses:

ST-6A IgG GMC (N = 1290)	Pre-Vac	Post-Vac	GMFR	ST-6A OPA GMT (N = 386)	Pre-Vac	Post-Vac	GMFR
BE-PCV-14 (n = 645)	0.448	1.075	2.4	BE-PCV14 (n = 192)	16.59	134.50	7.64
PCV-13 (n = 645)	0.446	1.906	4.27	PCV-13 (n = 194)	19.20	690.63	37.35

examination results did not indicate any safety issues of concern. Overall, the safety profile of BE-PCV-14 was comparable to the control vaccine PCV-13 in terms of AE rates, related AE rates and medically attended AEs and found to be safe and well tolerated.

4. Discussion

In this phase-3 trial, immunogenic non-inferiority, safety, and reactogenicity of a candidate BE-PCV-14 vaccine administered to 6–8 weeks old healthy infants in 6–10–14 weeks dosing schedule was evaluated. Robust and functional immune responses were elicited by BE-PCV-14 one month after the third vaccination dose. Primary basis for NI comparison and licensure decisions were based on reaching IgG concentrations ≥ 0.35 $\mu\text{g/ml}$ for all vaccine specific serotypes in the new vaccine and the licensed comparator. In accordance with WHO TRS (Technical Report Series) 977 guidelines [23], all 12 common serotypes in BE-PCV-14 met the non-inferiority criteria with those serotypes in PCV-13 in terms of both seroconversion rates (percentage of subjects with IgG ≥ 0.35 $\mu\text{g/ml}$) and GMC ratios (≥ 0.5). Both BE-PCV-14 and PCV-13 generated comparable functional immune responses (OPA titres), similar to IgG response.

Table 5
Overview of AEs- solicited and unsolicited AEs.

Category N1 (%) [95 % CI] n	BE-PCV-14 (N = 645)	PCV-13 (N = 645)
Overall AEs	175 (27.1 %) [23.73:30.74] 355	178 (27.6 %) [24.18:31.22] 353
Serious AEs	0 (0.0 %) [NE] 0	0 (0.0 %) [NE] 0
Severe AEs	0 (0.0 %) [NE] 0	0 (0.0 %) [NE] 0
Local AEs	137 (21.2 %) [18.14:24.60] 235	144 (22.3 %) [19.17:25.74] 251
Systemic AEs	70 (10.9 %) [8.56:13.51] 118	63 (9.8 %) [7.59:12.32] 101
Medically Attended AEs	0 (0.0 %) [NE] 0	0 (0.0 %) [NE] 0
AEs within 60 min postvac	4 (0.6 %) [0.17:1.58] 6	4 (0.6 %) [0.17:1.58] 5
AEs within 7 days postvac	174 (27.0 %) [23.59:30.58] 354	178 (27.6 %) [24.18:31.22] 352
AEs between Day 0 to Day 28	115 (17.8 %) [14.95:21.01] 207	109 (16.9 %) [14.09:20.02] 190
AEs between Day 28 to Day 56	61 (9.5 %) [7.31:11.98] 93	73 (11.3 %) [8.98:14.02] 110
AEs between Day 56 to Day 86	46 (7.1 %) [5.27:9.40] 55	41 (6.4 %) [4.60:8.52] 53
Any Solicited AEs	174 (27.0 %) [23.59:30.58] 353	178 (27.6 %) [24.18:31.22] 352
Any Unsolicited AEs	2 (0.3 %) [0.04:1.12] 2	1(0.2 %) [0.00:0.86] 1

Note: Percentages were calculated using respective column header group count as denominator.

95% CI was calculated by Clopper-Pearson Method.

N₁: Subject Count, N: Sample Size, n: Event Count.

General Note:

- All AE's were represented as: Subject count (Percentage of subjects) [95% CI] Event Count.
- Solicited Local and Systemic AEs were recored during 7 days (Day 0 – Day 6) after the vaccination. • Unsolicited adverse event reported at any time, until 28 days after the vaccination.

One of the secondary objectives was to evaluate the safety profile of the BE-PCV-14. Overall AE rates, related AE rates and medically attended AEs that were reported in BE-PCV-14 was comparable to PCV-13. The majority of reported AEs were mild in nature and vaccine related, with no reports of serious AEs in either treatment groups. The safety profile of BE-PCV-14 was similar to PCV-13 (also PCV-7), for which there is more than twenty years of clinical experience, pointing towards the high likelihood of continued safety of BE-PCV-14. In the current study, we also evaluated the cross-protection offered by serotype 6B against serotype 6A in BE-PCV-14 vaccinated group. Serotype 6A and 6B have high structural similarity of polysaccharide [24,25]. Therefore, it was expected that the immune responses induced by serotype 6B would provide at least partial cross-protection against serotype 6A in humans. An earlier study reported that PCV-13 elicits cross-functional opsonophagocytic killing responses in humans to serotypes not directly covered by the vaccine [26]. In this study, cross-protection was assessed by a ≥ 2 -fold increase in the 6A specific IgG antibody concentrations and the OPA antibody titers above the pre-vaccination levels. We found 2.4-fold and 7.64-fold increase in IgG GMC and OPA titers respectively, against serotype 6A in BE-PCV-14 vaccinated subjects, suggesting that BE-PCV-14 vaccine can offer protection against serotype 6A even though this serotype antigen is not present in BE-PCV-14. The serotype 6A cross protective functional immune response elicited by BE-PCV-14 is notably robust and strong when compared to another 10-valent PCV as described elsewhere [27]. This clearly indicates that serotype 6B in BE-PCV-14 affords significant cross protection to serotype 6A. Our results are consistent with previous studies, where it was shown that 23-valent polysaccharide vaccine lacking serotype 6A and carrying serotype 6B elicited cross-functional immune responses against serotype 6A [28,29].

Vaccine effectiveness (VE) against IPD was established in pediatric population for PCV-13 specific serotypes and it was found to be around 55 % [30]. Breakthrough IPD was mainly caused by serotype 3 and 19A

Table 6
Summary of AEs by SOC and PT.

SOC/PT,N1 (%) [95 % CI] n	BE-PCV-14 (N = 645)	PCV-13 (N = 645)
Overall	175 (27.1 %) [23.73:30.74] 355	178 (27.6 %) [24.18:31.22] 353
Gastrointestinal disorders	9 (1.4 %) [0.64:2.63] 12	5 (0.8 %) [0.25:1.80] 8
Abdominal pain	1 (0.2 %) [0.00:0.86] 1	0 (0.0 %) [NE] 0
Diarrhoea	6 (0.9 %) [0.34:2.01] 8	2 (0.3 %) [0.04:1.12] 3
Vomiting	3 (0.5 %) [0.10:1.35] 3	4 (0.6 %) [0.17:1.58] 5
General disorders and administration site conditions	169 (26.2 %) [22.85:29.78] 329	173 (26.8 %) [23.44:30.42] 333
Injection site erythema	33 (5.1 %) [3.55:7.11] 38	36 (5.6 %) [3.94:7.64] 44
Injection site induration	9 (1.4 %) [0.64:2.63] 13	13 (2.0 %) [1.08:3.42] 18
Injection site pain	93 (14.4 %) [11.80:17.37] 133	98 (15.2 %) [12.51:18.20] 131
Injection site swelling	44 (6.8 %) [5.00:9.05] 51	48 (7.4 %) [5.54:9.75] 58
Irritability Postvaccinal	25 (3.9 %) [2.52:5.67] 41	21 (3.3 %) [2.03:4.93] 31
Pyrexia	42 (6.5 %) [4.73:8.70] 53	40 (6.2 %) [4.47:8.35] 51
Metabolism and nutrition disorders	8 (1.2 %) [0.54:2.43] 9	6 (0.9 %) [0.34:2.01] 10
Decreased appetite	8 (1.2 %) [0.54:2.43] 9	6 (0.9 %) [0.34:2.01] 10
Nervous system disorders	2 (0.3 %) [0.04:1.12] 4	1 (0.2 %) [0.00:0.86] 2
Somnolence	2 (0.3 %) [0.04:1.12] 4	1 (0.2 %) [0.00:0.86] 2
Respiratory, thoracic and mediastinal disorders	1 (0.2 %) [0.00:0.86] 1	0 (0.0 %) [NE] 0
Cough	1 (0.2 %) [0.00:0.86] 1	0 (0.0 %) [NE] 0

Note: Percentages were calculated using column header group count as denominator.
95% CI were calculated by Clopper -Pearson Method.
N1: Subject Count, N: Sample Size, n: Event Count.
General Note:• All AE's were represented as: Subject count (Percentage of subjects) [95% CI] Event Count.

in PCV-13 vaccinated children. Excluding serotype 3 and 19A, VE of PCV-13 against rest of the vaccine serotypes was reported to be 89 % [30]. BE-PCV-14 was non-inferior to PCV-13 for all shared serotypes and against 2 unique serotypes (22F and 33F) when compared against lowest performing serotype 3 from PCV-13. Importantly, 83.5 % of BE-PCV-14 vaccinated children seroconverted against serotype 3 while 77 % seroconverted to serotype 3 in PCV-13 group. Several studies worldwide have shown that substantial proportion of pneumococcal disease is caused by vaccine serotypes, not included in the existing pneumococcal vaccines, predominantly by serotype 22F (13.4 %) and 33F (21 %) [19,31,32]. Another 15 valent PCV carrying these two serotypes was shown to induce serotype 22F and 33F immune response when compared to PCV-13 yy [33]. Similarly, the present study also demonstrates that BE-PCV-14 elicited serotype specific (22F and 33F) superior immune responses (Fig. 2b).

BE-PCV-14 containing 12 common serotypes and 22F and 33F serotypes along with cross protection to serotype 6A can protect the susceptible population against pneumococcal disease and may also provide herd immunity.

One limitation of the current study is that it is conducted in only Indian infant population and need to conduct similar studies in other regions. However, based on the VE and post marketing experience with PCV-13 across different geographical regions, it is likely that BE-PCV-14 will have similar effectiveness across regions.

In conclusion, BE-PCV-14 was safe, well tolerated and showed

Table 7
Overview AEs by Severity & Causality.

Category N1 (%) [95 % CI] n	BE-PCV-14 (N = 645)	PCV-13 (N = 645)
Number of subjects with at least one AE	175 (27.1 %) [23.73:30.74] 355	178 (27.6 %) [24.18:31.22] 353
Number of subjects with at least one SAE	0 (0.0 %) [NE] 0	0 (0.0 %) [NE] 0
Number of subjects with at least one MAAE	0 (0.0 %) [NE] 0	0 (0.0 %) [NE] 0
Severity		
Mild	174 (27.0 %) [23.59:30.58] 343	177 (27.4 %) [24.03:31.06] 338
Moderate	6 (0.9 %) [0.34:2.01] 12	10 (1.6 %) [0.75:2.83] 15
Severe	0 (0.0 %) [NE] 0	0 (0.0 %) [NE] 0
Life-threatening	0 (0.0 %) [NE] 0	0 (0.0 %) [NE] 0
Causality		
Certain	61 (9.5 %) [7.31:11.98] 79	65 (10.1 %) [7.86:12.66] 79
Probable	73 (11.3 %) [8.98:14.02] 84	68 (10.5 %) [8.28:13.17] 77
Possible	48 (7.4 %) [5.54:9.75] 185	53 (8.2 %) [6.22:10.61] 194
Unlikely	5 (0.8 %) [0.25:1.80] 6	3 (0.5 %) [0.10:1.35] 3
Unrelated	1 (0.2 %) [0.00:0.86] 1	0 (0.0 %) [NE] 0
Unclassifiable	0 (0.0 %) [NE] 0	0 (0.0 %) [NE] 0
Related (Certain/Probable/Possible)	173 (26.8 %) [23.44:30.42] 348	177 (27.4 %) [24.03:31.06] 350
Unrelated (Unlikely/Unrelated/Unclassifiable)	5 (0.8 %) [0.25:1.80] 7	3 (0.5 %) [0.10:1.35] 3
Number of subjects discontinued due to AE	0 (0.0 %) [NE] 0	0 (0.0 %) [NE] 0

comparable safety profile to that of PCV-13. BE-PCV-14 elicited immune responses non-inferior to PCV-13 in infants for all 12 common serotypes. Statistical NI was also met for two new serotypes 22F and 33F in BE-PCV-14 vaccine. BE-PCV-14 also induced cross protective antibodies against serotype 6A. These data support BE-PCV-14 protection against 14 pneumococcal serotypes and potential benefit against serotype 6A.

CRedit authorship contribution statement

Ramesh V. Matur: Conceptualization, Funding acquisition, Data curation, Writing, reviewing and editing the manuscript. **Subhash Thuluva:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Formal analysis, Data curation, Conceptualization. **Subbareddy Gunneri:** Writing – review & editing, Resources, Project administration, Methodology, Formal analysis, Data curation. **Vijay Yerroju:** Data Curation, Reviewing and editing the manuscript, formal analysis of the data. **Rammohan reddy Mogulla:** Data Curation, Reviewing and editing the manuscript, formal analysis of the data. **Kamal Thammireddy:** Writing – review & editing, Visualization, Validation, Methodology. **Piyush Paliwal:** Writing – review & editing, Validation, Formal analysis, Data curation. **Niranjana S. Mahantshetty:** Data Curation, Reviewing and editing the manuscript, formal analysis of the data, Validation and Methodology. **Mandyam Dhati Ravi:** Writing – review & editing, Supervision, Resources, Methodology, Investigation. **Savita Verma:** Writing – review & editing, Supervision, Resources, Methodology, Investigation. **Jai Prakash Narayan:** Data Curation, Reviewing and editing the manuscript, formal analysis of the data, Validation and Methodology.

Declaration of competing interest

The authors declare the following financial interests/personal

relationships which may be considered as potential competing interests: 'RVM, ST, SG, VY, RM, KT and PP are employees of Biological E Limited and they do not have any stock options or incentives. All the other participating authors declare no conflicts of interest.'

Data availability

Data will be made available on request.

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Data sharing agreement

Study data presented in the manuscript can be made available upon request and addressed to the corresponding author Dr. Ramesh V Matur at his email: Ramesh.Matur@biologicale.com.

Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.vaccine.2024.03.056>.

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